

Transition to the Open Complex (Simplified)

1. **Closed Complex:**
 - RNA polymerase binds promoter DNA, but DNA stays double-stranded.
 - This binding is **reversible**.
2. **Open Complex:**
 - RNA polymerase engages more tightly.
 - DNA “melts” (strands separate) at positions **–11 to +2**.
 - This is called **isomerization** (especially in bacteria with $\sigma 70$).
 - No ATP is needed; it happens spontaneously due to favorable DNA-enzyme interactions.
3. **How Melting Happens:**
 - Two bases in the –10 region of the **non-template strand** flip out and bind to σ protein pockets.
 - Stabilizes single-stranded DNA → promoter melts.
 - Once open, the complex is **irreversible** → transcription will start.
4. **RNA Polymerase Structure in Open Complex:**
 - **Active site:** at the base of the enzyme pincers (β and β' subunits).
 - **Five channels:**
 1. **NTP channel:** entry of ribonucleotides
 2. **RNA exit channel:** growing RNA leaves
 3. **Downstream DNA channel:** dsDNA enters
 4. **Template strand (T) channel:** template DNA path
 5. **Non-template strand (NT) channel:** non-template DNA path
5. **DNA Path:**
 - Downstream DNA enters double-stranded.
 - DNA separates at +3.
 - Template strand goes through T channel; non-template through NT channel.
 - Double helix reforms upstream at –11.
6. **Structural Changes in RNA Polymerase:**
 - **Pincers clamp** downstream DNA tightly.
 - **σ Region 1.1** moves out of the active site (50 Å shift) to let DNA in.
 - Acts as a **DNA mimic** (negatively charged like DNA).
 - Active site cleft is **positively charged**, helping DNA bind.

✓ Key Points to Remember:

- Closed → open = DNA melting + enzyme conformational change.
- No ATP needed.
- Open complex = irreversible → ensures transcription starts.

σ Initial Transcription (Simplified)

1. **Primer Not Needed:**
 - RNA polymerase can start a new RNA chain directly on the DNA template.
2. **Requirements:**
 - DNA template must fit correctly in the enzyme’s active site.
 - The first ribonucleotide (usually A) must align precisely with the template.
 - The second ribonucleotide must also be positioned correctly for the bond to form.
3. **Challenges:**
 - Starting with A is tricky because A-T base pairs have only **2 hydrogen bonds**, making the first bond less stable.
 - RNA polymerase must hold DNA and the first two nucleotides rigidly for polymerization.

4. **Abortive Synthesis:**
 - Initially, RNA polymerase makes **short RNAs (<10 nucleotides)** and releases them.
 - Only after this “trial” phase does it leave the promoter and enter **elongation**.
5. **How RNA Polymerase Moves Initially:**
 - Three proposed models explain the movement during these early cycles:
 1. **Transient Excursion:** Polymerase moves forward a little, makes a short RNA, then returns to promoter.
 2. **Inchworming:** A flexible module of the polymerase moves forward to synthesize RNA, then retracts.
 3. **Scrunching (Correct Model):** Polymerase stays **stationary** at the promoter, unwinds **downstream DNA**, and pulls it into itself, forming single-stranded loops (“scrunches”).

Elongating RNA Polymerase (Simplified)

1. **Role:**
 - RNA polymerase moves along DNA, **synthesizing RNA** and **proofreading** it.
2. **DNA Path:**
 - Double-stranded DNA enters the enzyme.
 - Strands separate at the active site:
 - **Template strand** guides RNA synthesis.
 - **Non-template strand** exits through a separate channel.
 - DNA reforms into a double helix behind the enzyme.
3. **RNA Synthesis:**
 - Ribonucleotides enter the active site and are added **one at a time** to the growing RNA.
 - Only **8–9 nucleotides** remain paired with DNA; the rest exit via the RNA channel.
4. **Movement (Stepping Mechanism):**
 - Polymerase acts like a **molecular motor**, advancing **one base pair per nucleotide added**.
 - The **transcription bubble** (unwound DNA) stays **constant in size**.
5. **Proofreading:**
 - **Pyrophosphoryl editing:** Removes wrong nucleotide using PP_i, then adds the correct one.
 - **Hydrolytic editing:** Polymerase backtracks, cuts out the error, and resumes synthesis.
6. **Support Proteins:**
 - **Nus proteins** join during elongation, helping RNA polymerase synthesize RNA efficiently and terminate properly.

✓ Key Points:

- RNA polymerase moves along DNA, unwinding ahead and rewinding behind.
- RNA grows one nucleotide at a time, with constant proofreading.
- The enzyme is **processive**, meaning it stays on DNA for long stretches without falling off.

Termination of Transcription (Simplified)

Purpose:

- Stops RNA synthesis at the end of a gene.
- Can also happen due to damaged DNA or other blocks.

Two Types in Bacteria:

1. **Rho-dependent termination**
 - Needs **Rho protein**.

- Rho is a **ring-shaped protein** that binds RNA at **rut sites** (C-rich, unstructured regions).
- Rho uses **ATP** to help terminate transcription.

How Rho works (possible models):

- **Pushes RNA polymerase** forward → polymerase falls off.
- **Pulls RNA** out of polymerase.
- **Changes polymerase shape** → stalling → termination.

Important:

- Rho usually acts **after transcription is done** for genes not being translated.
- Transcription and translation are coupled in bacteria; Rho avoids transcripts actively being translated.

2. Rho-independent (intrinsic) termination

- Does **not need Rho** or other factors.
- Relies on **DNA sequence elements**:
 - **Inverted repeat (~20 bp)** → forms a **stem-loop (hairpin)** in RNA.
 - **Followed by ~8 A:T base pairs.**

How it works:

- Hairpin forms in RNA → disrupts RNA polymerase.
- A:U pairs are weak → RNA easily separates from DNA → transcription stops.

Key Points:

- Termination ensures RNA polymerase **releases RNA and detaches from DNA**.
- Rho-dependent uses a protein to pull or push RNA polymerase.
- Rho-independent uses **RNA folding (hairpin)** to stop polymerase.
- Both mechanisms prevent errors and maintain proper gene expression.

RNA Splicing (Simplified)

1. Genes and Coding

- Genes have **coding sequences (exons)** that specify amino acids.
- In **prokaryotes**, coding sequences are usually continuous.
- In **eukaryotes**, coding sequences are **interrupted by non-coding sequences (introns)**.

2. Exons and Introns

- **Exons**: coding parts of the gene.
- **Introns**: non-coding parts that interrupt exons.
- A gene may have **1 intron** (like yeast) or **hundreds** (like Titin in humans).
- Introns are often **much longer** than exons.

Example:

- Dihydrofolate reductase gene → 31 kb gene, only 2 kb is coding (exons) → ~10% coding.

3. Primary Transcript (pre-mRNA)

- Eukaryotic genes are transcribed into **pre-mRNA** containing **both exons and introns**.
- Pre-mRNA can be **very long** due to introns.

4. RNA Splicing Process

- **Goal:** Remove introns and join exons → create **mature mRNA**.
- **Precision is crucial:** skipping or adding even 1 nucleotide changes the **reading frame** → wrong protein.

5. Alternative Splicing

- Some pre-mRNAs can be spliced in **different ways** → different combinations of exons.
- Produces **multiple proteins (isoforms)** from a single gene.
- In humans, ~90% of genes undergo **alternative splicing** → increases protein diversity.

✓ Key points to remember:

1. Exons = coding, Introns = non-coding
2. Pre-mRNA = exons + introns
3. Splicing removes introns → mature mRNA
4. Alternative splicing → multiple protein forms from 1 gene

Splicing Site Determination (Simplified)

1. How does the cell know where to splice?

- Introns and exons are recognized by **specific sequences in the pre-mRNA**.
- These sequences tell the **spliceosome** exactly **where to cut and join**.

2. Key sequences in introns

1. **5' splice site (donor site)** – at the **start of the intron**
 - Usually begins with **GU**
2. **3' splice site (acceptor site)** – at the **end of the intron**
 - Usually ends with **AG**
3. **Branchpoint site** – located **inside the intron**, near the 3' end
 - Has a conserved **A nucleotide**
 - Followed by a **polypyrimidine tract (Py tract)**

3. How splicing happens

- The **spliceosome** recognizes these sequences: GU, A, and AG.
- Cuts at the **5' and 3' splice sites**.
- Joins the exons together with **high precision**.

4. Why introns have these sequences

- Exons are constrained by **coding for proteins**, so introns carry the signals (GU, AG, branchpoint) for **splicing**.

Removal of Introns (Simplified)

1. Two-step chemical reaction

- Introns are removed by **two transesterification reactions**.
- These reactions **break and reform phosphodiester bonds** in the pre-mRNA.

2. First reaction

- The **2'-OH of the conserved A** at the **branch point** attacks the **phosphate at the 5' splice site**.
- This **cuts the 5' end of the intron**.
- The 5' end of the intron attaches to the **branchpoint A**, forming a **looped structure called a lariat**.

3. Second reaction

- The **5' exon** now attacks the **3' splice site**.
- This **joins the 5' and 3' exons together** → actual splicing occurs.
- The intron is **released in a lariat form**.

4. Energy and direction

- **No net energy** is needed to break and form bonds.
- **ATP is required** to assemble and operate the **spliceosome**.
- Splicing is **unidirectional** in cells; it normally does not go backward.

The Spliceosome (Simplified)

1. What it is

- Huge molecular machine that **removes introns** from pre-mRNA.
- Made of **~150 proteins + 5 RNAs** (U1, U2, U4, U5, U6).

- Similar in size to a ribosome.
- Uses **ATP** to work, but the chemical splicing itself doesn't need energy.

2. RNA does most of the work

- The **RNAs (snRNAs)** do most of the splicing and help **recognize intron/exon borders**.
- Proteins mostly **stabilize the complex and help with rearrangements**.

3. snRNPs ("snurps")

- snRNAs + proteins = **small nuclear ribonucleoproteins (snRNPs)**.
- Form the **spliceosome**.
- Roles of snRNPs:
 1. Recognize **5' splice site** and **branch site**
 2. Bring **splice sites together**
 3. **Catalyze splicing reactions**

4. How it works

- **U1 snRNA** binds the **5' splice site**.
- **U2 snRNA** binds the **branch point**.
- **U2 + U6 snRNAs** bring the 5' splice site and branch site together → positions them for the two-step reaction.

5. Other proteins

- **U2AF** → recognizes polypyrimidine tract (near 3' splice site).
- **BBP/SF1** → binds branch point initially, then replaced by U2 snRNP.
- **DEAD-box helicases** → use ATP to rearrange RNA-RNA interactions, remove spliced mRNA, and disassemble spliceosome.
- **RNA-annealing factors** → help load snRNPs onto pre-mRNA.

The Splicing Pathway (Simplified)

1. Early Recognition (E Complex)

- **U1 snRNP** binds the **5' splice site**.
- **U2AF** (two subunits: 65 + 35) binds the **polypyrimidine tract** and **3' splice site**.
- **BBP (branchpoint binding protein)** binds the **branch site**.
- This combination = **Early (E) complex**.

2. A Complex Formation

- **U2 snRNP** replaces BBP at the branch site.
- The branch **A nucleotide bulges out**, ready to attack the 5' splice site.
- This is now called the **A complex**.

3. B Complex Formation (Tri-snRNP)

- **U4, U5, U6 snRNPs** join the A complex → called **tri-snRNP**.
- U1 leaves → U6 takes its place at the **5' splice site**.
- This creates the **B complex**, fully assembled but not yet active.

4. C Complex Formation (Active Site)

- **U4 leaves**, allowing **U6 and U2** to interact → forms the **active site**.
- Active site = mostly **RNA**, only formed at this stage → ensures **precision**.
- Positions the **5' splice site** and **branch site** for the first reaction.

5. Catalysis

- **Step 1:** Branch **A attacks 5' splice site** → forms **lariat**.
- **Step 2:** 5' exon attacks **3' splice site** → joins the exons → removes intron.
- **U5 snRNP** helps bring the two exons together.

6. Release & Recycling

- Mature **mRNA** is released.
- **snRNPs** are recycled.
- The **lariat intron** is degraded.

✓ Key Points

- Spliceosome = stepwise assembly: **E → A → B → C complex**.
- Active site forms **only at the right time** → prevents errors.
- Catalysis = **two transesterification reactions**.
- **Precision** is ensured by RNA-RNA interactions and snRNP arrangement.

Self-Splicing Introns (Simplified)

Key Idea

- Some introns can **splice themselves out** without any proteins or spliceosome.
- RNA itself acts as a **catalyst**.
- There are **three types of splicing in cells**:
 1. **Nuclear pre-mRNA** (spliceosome-mediated)

2. **Group II introns** (self-splicing)
3. **Group I introns** (self-splicing)

What “Self-Splicing” Means

- The intron **folds into a 3D shape** within the RNA.
- It catalyzes **its own removal**.
- Works **in a test tube** without proteins or other RNAs.
- Usually only does **one round** of splicing → not a typical enzyme.

Group II Introns (Example)

1. A branch **A nucleotide** in the intron attacks the **5' splice site** (between exon and intron).
 - This forms a **lariat** structure.
2. The **3'-OH of the 5' exon** attacks the **3' splice site**.
 - This **joins the exons** and releases the intron as a lariat.

Genetic Code

- **Codon:** 3 nucleotides in mRNA = 1 amino acid (AA).
- Complementary and **antiparallel** to the DNA template.
- **Features:**
 - Universal (same in almost all organisms)
 - Redundant (multiple codons for some AAs)
 - Not ambiguous (each codon codes for only one AA)
- **Special codons:**
 - **Start codon:** signals beginning of translation
 - **Stop codons:** signal termination

The Adapter: tRNA

- tRNA **translates nucleotide language into amino acid language**.
- **Functions:**
 1. Carries an amino acid.
 2. Binds to mRNA codons via anticodon.
 3. Binds ribosome to help form polypeptide.
- **Structure:**
 - 75–80 nucleotides long
 - Internal hydrogen bonding → forms cloverleaf structure
 - **3' end (CCA):** attaches to amino acid
 - **Anticodon loop:** pairs antiparallel with mRNA codon

Translation Overview

- Translation: **mRNA** → **protein**
- Highly conserved and energy-intensive process
 - Up to 80% of bacterial cell energy can go into protein synthesis
- Requires >100 proteins and RNAs for a single protein

Why tRNA is Needed

- Amino acids **cannot directly recognize codons** in RNA.
- tRNA acts as an **adapter** that links codons to specific amino acids.
- First proposed by Francis Crick (1955)
- Confirmed by Zamecnik & Hoagland (1957): amino acids attach to tRNA before protein synthesis

Translation Machinery

Four main components:

1. **mRNA:** provides the codon sequence (template)
2. **tRNA:** carries amino acids and matches codons via anticodons
3. **Aminoacyl-tRNA synthetases:** attach amino acids to correct tRNAs
4. **Ribosome:** RNA + protein complex
 - Coordinates codon-anticodon recognition
 - Catalyzes peptide bond formation

Summary of Process

1. mRNA codons are read by tRNA anticodons.
2. tRNAs bring the correct amino acids.
3. Ribosome links amino acids into a growing polypeptide chain.
4. Result → protein translated according to mRNA instructions.

Messenger RNA (mRNA) and Open Reading Frames (ORFs)

- **ORF (Open Reading Frame):**
 - A **contiguous, non-overlapping string of codons** in mRNA that codes for a protein.
 - Each ORF specifies a **single protein**.
 - ORF starts with a **start codon** and ends with a **stop codon**.
 - ORF boundaries are **internal to the mRNA**; not necessarily the ends of the mRNA itself.
- **Start Codon:**
 - **Eukaryotes:** AUG only
 - **Bacteria:** Usually AUG, sometimes GUG or UUG
 - Functions:
 1. Specifies first amino acid in the polypeptide
 2. Sets the **reading frame** for translation
- **Stop Codons:** UAG, UGA, UAA
 - Define the end of the ORF and signal **termination of translation**
- **Open Reading Frame Concept:**
 - A stretch of codons “open” for translation without a stop codon

- Each stretch could theoretically be read in 3 different frames; the start codon **fixes the correct frame**

ORFs in Eukaryotes vs Prokaryotes

Feature	Eukaryotic mRNA	Prokaryotic mRNA
ORFs per mRNA	Usually 1 (monocistronic)	Often 2 or more (polycistronic)
Function of multiple ORFs	Not common	Usually encode related proteins (e.g., same pathway)

Prokaryotic mRNAs and Ribosome Binding

- **Ribosome Binding Site (RBS):**
 - Short sequence upstream of the start codon
 - Also called **Shine-Dalgarno sequence**
 - Complementary to **16S rRNA** (3' end) in the ribosome
 - Typical RBS sequence: AGGAGG
 - Spacing between RBS and start codon influences translation efficiency
- **Translational Coupling:**
 - Occurs in polycistronic mRNAs
 - Downstream ORFs often **overlap** upstream ORFs
 - Ribosome finishing upstream ORF can start translating downstream ORF directly
 - Mutation introducing a premature stop in upstream ORF can **prevent translation of downstream ORF**

Eukaryotic mRNAs and Translation Initiation

1. 5' Cap

- A **methylated guanine nucleotide** attached to the 5' end of mRNA via a **5'-to-5' triphosphate linkage**.
- Functions:
 1. **Recruits the ribosome** to the mRNA.
 2. Guides **ribosome scanning** along the mRNA in the **5' → 3' direction** to locate the start codon (AUG).
- Essential for **translation initiation** and also **protects mRNA** from degradation.

2. Kozak Sequence

- A consensus sequence around the start codon:
5'-G/ANNAUGG-3'
- Features:
 - Purine three bases upstream of AUG
 - Guanine immediately downstream
- Function: Enhances **translation efficiency** by improving initiator tRNA recognition.
- Not all eukaryotic mRNAs have a perfect Kozak sequence, but its presence boosts translation.

3. Poly-A Tail

- A string of adenine nucleotides added enzymatically to the **3' end** by **poly-A polymerase**.
- Functions:
 1. Enhances **translation efficiency** by promoting recruitment of initiation factors.
 2. Protects mRNA from **rapid degradation**.
- Works together with the 5' cap to stabilize the mRNA and support translation.

Summary of Eukaryotic Translation Initiation

1. Ribosome binds to the **5' cap**.
2. Ribosome **scans along the mRNA** to find the AUG start codon (helped by the Kozak sequence if present).
3. Poly-A tail at 3' end boosts efficiency and protects mRNA.
4. Translation starts, producing the protein encoded by the ORF.

Transfer RNA (tRNA) – The Adaptor Molecule

Function

- tRNAs are adaptors that **translate mRNA codons into amino acids** during protein synthesis.
- Each tRNA is specific for:
 1. **One amino acid** (cognate amino acid)
 2. **One or more codons** in mRNA

General Features

- Length: **75–95 nucleotides**
- Ends at **3' terminus with CCA**, the site for amino acid attachment.
- Contains **modified bases** for stability and function:
 - **Pseudouridine (ΨU)**: Isomer of uridine; improves tRNA function.
 - **Dihydrouridine (D)**: Derived from uridine; contributes to structure.
 - **Hypoxanthine, methylguanine, thymine** – help in codon recognition and stability.

Secondary Structure: Cloverleaf

tRNA folds into a conserved **cloverleaf pattern** with stems and loops:

1. **Acceptor Stem**
 - Formed by base pairing between 5' and 3' ends.
 - Site of **amino acid attachment**.
2. **Ψ (Pseudouridine) Loop**
 - Contains modified base ΨU.
 - Often has sequence **5'-TCUCG-3'**.
3. **D Loop**
 - Contains **dihydrouridine** residues.
 - Contributes to tRNA folding and recognition.
4. **Anticodon Loop**
 - Contains the **anticodon**, a 3-nucleotide sequence complementary to the mRNA codon.
 - Responsible for **codon recognition**.
5. **Variable Loop**

- Between anticodon and Ψ loop.
- Varies in length (**3–21 nucleotides**).

Key Points

- **Modified bases** are not essential but **enhance tRNA function** and translation efficiency.
- tRNA serves as the **physical link** between mRNA codons and the growing polypeptide chain.
- The **anticodon loop** ensures proper reading of the mRNA and accurate incorporation of amino acids

Charging tRNAs with Amino Acids (Aminoacylation)

Definitions

- **Charged tRNA:** tRNA with an amino acid attached.
- **Uncharged tRNA:** tRNA without an amino acid.
- **Acyl linkage:** High-energy bond formed between amino acid and tRNA.

Significance:

- The energy stored in this bond drives **peptide bond formation** during translation.

Step 1: Adenylation

1. The **amino acid reacts with ATP**, forming an **aminoacyl-AMP intermediate**.
2. **Pyrophosphate (PPi)** is released.
3. The amino acid is now linked to AMP via a **high-energy ester bond**.

Note: This step is called **adenylation** (transfer of AMP, not adenine).
Pyrophosphate hydrolysis drives the reaction forward.

Step 2: tRNA Charging

1. The **adenylylated amino acid** reacts with the **3' end of the tRNA**.
2. The amino acid is transferred to either the **2'-OH or 3'-OH** of the terminal adenosine of tRNA.
3. **AMP is released**, and the tRNA is now **charged**.
4. After release, the amino acid **equilibrates** between 2'-OH and 3'-OH positions.

Classes of Aminoacyl-tRNA Synthetases

1. **Class I enzymes**
 - Attach amino acid to **2'-OH** of tRNA.
 - Usually **monomeric**.
2. **Class II enzymes**
 - Attach amino acid to **3'-OH** of tRNA.
 - Usually **dimeric or tetrameric**.

Specificity

- Each **amino acid has a dedicated synthetase**.
- Many amino acids have multiple codons → one synthetase may recognize **multiple tRNAs** (isoaccepting tRNAs).
- **One synthetase per amino acid** ensures **accuracy** in translation.

Key Points

- The acyl linkage is **high-energy**, coupling amino acid attachment to peptide bond formation.
- tRNA charging is essential for **translating the genetic code** into a protein.
- Fidelity of synthetases ensures **correct amino acid incorporation**.

Peptide Bond Formation

Direction of Synthesis

- Proteins are synthesized **from the amino (N) terminus to the carboxyl (C) terminus**.
- Each new amino acid is added to the **C-terminal end** of the growing polypeptide chain.

Key Players

1. **Aminoacyl-tRNA** – tRNA carrying the **incoming amino acid**.
2. **Peptidyl-tRNA** – tRNA carrying the **growing polypeptide chain**.
3. **Ribosome** – catalyzes the formation of the **peptide bond**.

Mechanism

1. Both the **aminoacyl-tRNA** and **peptidyl-tRNA** are bound at the ribosome's **3' ends**.
2. The ribosome **positions the tRNAs** so that:
 - The amino group of the incoming amino acid can attack the **carbonyl group of the polypeptide chain** attached to the peptidyl-tRNA.
3. **Nucleophilic attack occurs**:
 - A new **peptide bond** is formed between the amino acids.
 - The polypeptide chain is **transferred** from the peptidyl-tRNA to the aminoacyl-tRNA.

This reaction is called the **peptidyl transferase reaction**.

Energy Considerations

- No ATP or GTP is **directly hydrolyzed** during peptide bond formation.
- The **energy comes from the high-energy bond** formed when the amino acid was attached to its tRNA during tRNA charging.

- This means the peptide bond formation is **driven by the energy stored in aminoacyl-tRNAs**.

Consequences of This Mechanism

1. **N-terminus first:** Amino acids are added sequentially from **N-terminal to C-terminal**.
2. **Polypeptide transfer:** The growing chain is transferred from the **peptidyl-tRNA** to the **aminoacyl-tRNA**, enabling chain elongation.

Ribosome Structure and Function

- Ribosomes are **molecular machines** made of **RNA and proteins**.
- They have **two subunits**:
 - **Large subunit** → contains the **peptidyl transferase center** (PTC), where peptide bonds are formed.
 - **Small subunit** → contains the **decoding center**, where tRNAs "read" codons on the mRNA.

Polymers Entering and Exiting the Ribosome

- mRNA threads through the **decoding center** of the small subunit.
- The nascent polypeptide exits from the **peptidyl transferase center** of the large subunit.
- **Ribosomal tunnels** provide pathways for these polymers.

tRNA Binding Sites

- The ribosome can bind **at least two tRNAs simultaneously**, but actually contains **three binding sites**:
 1. **A-site (Aminoacyl site)**
 - Binds **aminoacyl-tRNA** (tRNA carrying a new amino acid).
 2. **P-site (Peptidyl site)**
 - Binds **peptidyl-tRNA** (tRNA holding the growing peptide chain).
 3. **E-site (Exit site)**
 - Binds **tRNA that has donated its amino acid** and is about to exit.

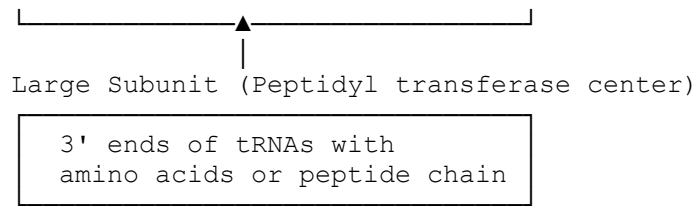
Spatial Organization

- Each tRNA binding site is **at the interface of the large and small subunits**.
- This allows tRNAs to **connect the peptidyl transferase center with the decoding center**.
- **Orientation of tRNAs**:
 - **3' ends** (where amino acids attach) → near the **large subunit**.
 - **Anticodon loops** (which pair with mRNA codons) → near the **small subunit**.

Summary Diagram (Mental Visualization)

Small Subunit (Decoding center)

Anticodon loops of tRNAs (bind mRNA codons)
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tRNA Sites: A → P → E

Initiation of Translation

Translation initiation is the process by which the ribosome is correctly positioned on the mRNA and the first tRNA is loaded to start protein synthesis. For initiation to occur, **three key events** must happen:

1. **Recruitment of the ribosome to the mRNA**
2. **Placement of a charged tRNA into the P-site of the ribosome**
3. **Precise positioning of the ribosome over the start codon** (establishes the reading frame for correct translation)

Step 1: Ribosome Assembly in Prokaryotes

- In prokaryotes, **ribosome assembly occurs one subunit at a time**:
 - **Small subunit** binds first to the mRNA.
 - This binding is guided by base-pairing between the **Ribosome Binding Site (RBS)** on the mRNA and **16S rRNA** of the small subunit.
- Proper RBS positioning ensures that the **start codon** (AUG or sometimes GUG) is in the **P-site** when the **large subunit joins**.

Step 2: Large Subunit Joins

- The **large subunit** binds **only after the small subunit is correctly positioned** on the mRNA.
- Many initiation steps happen **without the full ribosome**, emphasizing the importance of small subunit positioning.

Step 3: Initiator tRNA Binding

- Translation initiation is unique because the **tRNA enters the P-site first**, bypassing the A-site.
- This requires a **special initiator tRNA**:
 - Base-pairs with the **start codon** (AUG or GUG)
 - Initially charged with **methionine**, but then modified to **N-formylmethionine (fMet)** by **Met-tRNA transformylase**
 - This gives rise to **fMet-tRNA^{fMet}**

Step 4: Fate of N-Formyl Methionine

- Although N-formylmethionine starts most prokaryotic proteins:
 - **Deformylase** removes the formyl group during or after translation.
 - **Aminopeptidases** may remove the initial methionine and sometimes one or two additional amino acids.
- As a result, many mature proteins **do not start with methionine**.

Key Points to Remember

- **Start codon placement** = establishes reading frame.
- **Initiator tRNA (fMet-tRNA^{fMet})** = only tRNA to enter P-site first.
- **RBS-16S rRNA interaction** = critical for small subunit positioning.
- **Formyl group removal** = ensures proper mature protein sequence.

Prokaryotic Translation Initiation Factors

The initiation of translation is catalyzed by **three initiation factors: IF1, IF2, and IF3**. Each factor plays a **specific role** in preparing the small ribosomal subunit for translation.

1. IF1

- **Function:** Blocks the A-site of the small subunit to **prevent tRNAs from entering the A-site** prematurely.
- **Binding site:** Occupies the portion of the small subunit that will become the **A-site**.
- **Role:** Ensures that only the **P-site** can initially bind tRNA (the initiator tRNA).

2. IF2

- **Function:** A **GTPase** that facilitates binding of the **initiator tRNA (fMet-tRNA^{fMet})** to the small subunit.
- **Interactions:** Contacts the **small subunit, IF1, and fMet-tRNA**.
- **Role:**
 - Prevents other tRNAs from binding.
 - Acts as a **docking site for the large subunit**.
 - Stimulates **GTP hydrolysis** upon large subunit binding, which triggers IF release.
- **Binding site:** Reaches over the **A-site into the P-site** to hold fMet-tRNA^{fMet}.

3. IF3

- **Function:**
 - Binds to the **small subunit** to **prevent reassociation** with the large subunit.
 - Ensures a free small subunit is available for **new translation cycles**.
- **Role in recycling:** Helps dissociate the 70S ribosome after previous translation rounds.
- **Binding site:** Occupies the part of the small subunit that will become the **E-site**.

Effect of Initiation Factors on tRNA Binding

- With IF1, IF2, and IF3 bound:
 - **A-site** → blocked by IF1
 - **P-site** → free to bind **fMet-tRNA^{fMet}**
 - **E-site** → blocked by IF3
- This ensures only the **initiator tRNA** binds at the start of translation.

Sequence of Events During Initiation

1. **Small subunit + IFs bind mRNA and fMet-tRNA^{fMet}**
 - fMet-tRNA^{fMet} base-pairs with the start codon (AUG/GUG) in the P-site.
 - Binding is stabilized by IF2-GTP.
2. **Conformational change in the small subunit**
 - Occurs upon start codon recognition.
 - Leads to the **release of IF3**, allowing large subunit binding.
3. **Large subunit association**
 - IF2 acts as a docking site.
 - GTP hydrolysis by IF2 triggers **release of IF1 and IF2-GDP**.
4. **Result:**
 - Formation of a **70S initiation complex**.
 - **fMet-tRNA^{fMet} in P-site, empty A-site**, ready for the first elongator tRNA.

Key Points

- Only the **P-site is active initially**, thanks to the IFs.
- IF2 is central for **initiator tRNA binding and large subunit docking**.
- IF3 ensures **small subunit recycling** and proper start site selection.
- IF1 prevents premature **A-site occupation** by other tRNAs.

Translation Elongation

Once the ribosome is assembled with **fMet-tRNA^{fMet} in the P-site**, elongation begins. The elongation cycle repeats for each amino acid added to the growing polypeptide.

Three Key Events for Each Amino Acid Addition

1. **Aminoacyl-tRNA Selection**
 - The correct **aminoacyl-tRNA** enters the **A-site**, guided by **codon-anticodon base pairing**.
 - tRNAs are **not free-floating**; they are escorted by the elongation factor **EF-Tu** (bound to GTP).
 - EF-Tu binds the 3' end of the aminoacyl-tRNA, masking the amino acid to prevent premature peptide bond formation.
2. **Peptide Bond Formation**
 - The **peptidyl transferase center** of the large subunit catalyzes a **peptide bond** between:
 - The amino acid on the **A-site tRNA**, and
 - The growing peptide attached to the **P-site tRNA**.
 - Result: The **growing peptide chain is transferred to the tRNA in the A-site**.
3. **Translocation**
 - The **ribosome moves one codon** along the mRNA.
 - The **peptidyl-tRNA in the A-site shifts to the P-site**.

- The **A-site becomes empty** and ready for the next aminoacyl-tRNA.
- Precise movement is critical to **maintain the reading frame**.

Role of Elongation Factors

EF-Tu

- **Function:** Escorts aminoacyl-tRNAs to the A-site.
- **Mechanism:**
 1. Binds aminoacyl-tRNA only when **GTP is bound**.
 2. EF-Tu-GTP-tRNA complex enters the ribosome.
 3. Correct **codon-anticodon match** triggers **GTP hydrolysis**.
 4. EF-Tu-GDP releases the tRNA into the A-site.
- Ensures **accuracy of translation**; error rate $\approx 10^{-3}$ to 10^{-4} .

Factor Binding Center

- Domain on the **large subunit** that activates GTP hydrolysis by EF-Tu (also activates IF2 during initiation).
- Ensures that only **correctly matched tRNAs** are released from EF-Tu.

Accuracy of Elongation

- The **main determinant** of correct amino acid incorporation is **base pairing** between codon (mRNA) and anticodon (tRNA).
- **Mismatched tRNAs** are rarely allowed to participate, maintaining **high fidelity** of protein synthesis.

Key Points

- **Elongation is conserved** between prokaryotes and eukaryotes.
- Each cycle involves **A-site entry** → **peptide bond formation** → **translocation**.
- **EF-Tu-GTP hydrolysis** is central to both **accuracy and timing** of elongation.
- The **ribosome ensures correct reading frame** at every step.

The Ribosome Is a Ribozyme

The ribosome is **not just a scaffold** for protein synthesis—it **catalyzes peptide bond formation**, and this activity is carried out by **RNA**, not protein.

Peptidyl Transferase Activity

- **Peptide bond formation** occurs when the **aminoacyl-tRNA in the A-site** rotates into the **peptidyl transferase center (PTC)** of the large subunit.
- The reaction is **catalyzed by the 23S rRNA** of the large subunit.
- Evidence:
 - Large subunits stripped of nearly all proteins can still catalyze peptide bonds.

- High-resolution 3D structures show **no amino acid is within 18 Å of the active site**, far too distant to participate chemically.
→ Confirms the PTC is an **RNA enzyme (ribozyme)**.

Mechanism of Catalysis (Current Understanding)

1. **Substrate Positioning**
 - Base-pairing occurs between **23S rRNA** and the **CCA ends** of the tRNAs in the **A and P sites**.
 - This **correctly positions the α -amino group** of the aminoacyl-tRNA to attack the **carbonyl group** of the peptide attached to the P-site tRNA.
2. **Substrate Stabilization**
 - Interactions with 23S rRNA also **stabilize the aminoacyl-tRNA** in the PTC after it “accommodates” in the active site.
3. **Chemical Facilitation**
 - The ribosome likely **modifies the chemical environment** of the active site.
 - Proposed mechanism: **ribosomal nucleotides may accept a hydrogen from the α -amino group**, making the nitrogen a stronger **nucleophile**.
 - This is similar to how some protein enzymes **stimulate nucleophilic attack** on carbonyl groups.

Key Takeaways

- **Catalysis is RNA-mediated**, making the ribosome a **ribozyme**.
- Proteins in the ribosome are **structural/supportive**, not catalytic.
- Precise positioning and chemical activation of substrates enable **efficient peptide bond formation**.
- This mechanism explains the ribosome’s **remarkable speed and fidelity** in elongating polypeptides.

Translocation in the Ribosome

Translocation is the coordinated movement of tRNAs and mRNA within the ribosome after **peptide bond formation**, preparing the ribosome for the next round of elongation.

Step 1: Peptidyl Transfer and Hybrid State Formation

- After the **peptidyl transferase reaction**:
 - The **P-site tRNA** is now **deacylated** (no amino acid).
 - The **A-site tRNA** carries the **growing polypeptide**.
- **Preferred positions in the large subunit**:
 - **A-site tRNA 3' end** → **P-site of large subunit**
 - **P-site tRNA 3' end** → **E-site of large subunit**
- **Anticodon ends** remain in the **small subunit** (still bound to mRNA).
- This creates a “**hybrid state**”, where 3' ends of tRNAs have moved but anticodons have not.
- Associated with **counterclockwise rotation of the small subunit** relative to the large subunit, facilitating tRNA interactions with distinct binding sites in each subunit.

Step 2: EF-G–Mediated Completion of Translocation

- **EF-G**, bound to **GTP**, binds the ribosome in the **rotated, hybrid state**.
- EF-G contacts the **factor binding center** of the large subunit → **stimulates GTP hydrolysis**.

Consequences of EF-G–GTP hydrolysis:

1. **Unlocking the ribosome**
 - EF-G–GDP opens “gates” between the **A-, P-, and E-sites**, allowing tRNA movement.
2. **A-site competition**
 - EF-G–GDP occupies the **A-site of the decoding center**, competing with the tRNA.
 - This facilitates the **A-site tRNA moving into the P-site**.

Step 3: Return to the Locked State

- After translocation:
 - Small subunit rotates **clockwise back to starting position**.
 - Ribosome regains a **locked conformation**:
 - tRNAs tightly bound in the **P- and E-sites**.
 - mRNA codon correctly positioned in the decoding center.
 - Gates between A-, P-, and E-sites closed.
- **Result:**
 - **A-site is empty**, ready for the next aminoacyl-tRNA.
 - **P-site tRNA** carries the polypeptide.
 - **E-site tRNA** ready to exit.
 - **mRNA has shifted by 3 nucleotides**.

Key Points

- Translocation begins in the **large subunit** (hybrid state) before the small subunit moves.
- EF-G–GTP hydrolysis **drives the mechanical movement**.
- Small subunit rotation is essential for **synchronized tRNA-mRNA movement**.
- Ensures **correct reading frame** and **precision in elongation**.

Termination of Translation

Translation continues until a **stop codon (UAA, UAG, or UGA)** enters the **A-site**. Stop codons are **not recognized by tRNAs**, but by **protein release factors (RFs)** that trigger **polypeptide release**.

Release Factors

Two classes of release factors:

Class I RFs

- **Function:** Recognize stop codons and trigger **hydrolysis of the polypeptide from the P-site tRNA**.
- **Prokaryotes:**
 - **RF1:** recognizes **UAG**
 - **RF2:** recognizes **UGA**

- **Both RF1 and RF2** recognize UAA
- **Eukaryotes:**
 - **eRF1:** recognizes **all three stop codons**
- **Mechanism:**
 - Functionally **mimic a tRNA**.
 - Possess a **peptide anticodon** that interacts with the stop codon.
 - Contain a **conserved GGQ motif** (Gly-Gly-Gln) that is critical for polypeptide release at the **peptidyl transferase center**.
 - GGQ may directly catalyze hydrolysis or induce a conformational change in the PTC.

Class II RFs

- **Function:** Stimulate **dissociation of Class I RFs** from the ribosome after peptide release.
- **Prokaryotes: RF3**
- **Eukaryotes: eRF3**
- **Mechanism:**
 - GTP-binding proteins.
 - Unlike other GTPases (EF-G, IF2, EF-Tu), they have **higher affinity for GDP** in their free form.
 - RF3-GDP binds the ribosome **only in the presence of a Class I RF**.
 - After polypeptide release, RF3 exchanges GDP for GTP → stabilizes **rotated ribosome conformation**, displaces Class I RF, and binds the factor-binding center.
 - GTP hydrolysis leads to **release of RF3-GDP**, returning the ribosome to a **locked state** ready for recycling.

Recognition of Stop Codons

- Stop codon recognition is mediated by **protein-RNA interactions**.
- Experiments swapping small regions of RF1 and RF2 identified a **three-amino-acid sequence (peptide anticodon)** critical for codon specificity.
- RF1 binds to the **A-site**, positioning its peptide anticodon near the codon.

Summary of Termination Steps

1. **Stop codon enters A-site** → recognized by **Class I RF**.
2. **Polypeptide hydrolysis** from P-site tRNA occurs via the **GGQ motif** of Class I RF.
3. **Class II RF (RF3/eRF3)** binds → promotes **release of Class I RF**.
4. **Ribosome returns to locked state**, ready for **subunit recycling** or new translation cycle.

Key Points

- Stop codons **do not have corresponding tRNAs**.
- **Class I RFs** mimic tRNA and catalyze peptide release.
- **Class II RFs** use GTP hydrolysis to remove Class I RFs.
- Proper termination ensures **full-length, accurate protein synthesis** and ribosome recycling.

Protein Targeting

In Prokaryotes

- **Polyribosomes (Polysomes):**
 - Multiple ribosomes translate a single mRNA simultaneously.
- **Simultaneous Transcription and Translation:**
 - Translation begins even before transcription is complete.

In Eukaryotes

- **Compartmentalization:** Proteins are targeted to specific organelles.
- **Signal Sequence:**
 - Short stretch (~20 hydrophobic amino acids) at the **N-terminus**.
 - Directs the nascent polypeptide to the **rough ER**.
- **Mutations affecting signal sequences** can mislocalize proteins.

Mutations

Types of Mutations

1. **Somatic Mutations:**
 - Occur in body cells.
 - Passed to daughter cells after mitosis, **not inherited by offspring**.
2. **Germ-line Mutations:**
 - Occur in gamete-producing cells.
 - Passed to the next generation.

Based on DNA Changes

- **Point Mutations:** Single base change
 - **Silent:** No change in amino acid (redundancy in code)
 - **Missense:** One amino acid replaced by another (e.g., **Sickle-cell disease**, β -globin mutation)
 - **Nonsense:** Creates a stop codon $\rightarrow$ truncated protein
- **Frame-Shift Mutations:**
 - Insertion or deletion of a base $\rightarrow$ shifts reading frame.
- **Chromosomal Mutations:** Changes in **position/orientation** of DNA segments.

Impact

- Can be **beneficial** (increase diversity, drive evolution) or harmful (diseases).

Transcription Regulation

Gene Categories

1. **Housekeeping genes:** Expressed all the time in all cells (routine metabolism).
2. **Differentiation-specific genes:** Expressed when a cell enters a specific pathway.
3. **Cell-type-specific genes:** Always expressed in certain differentiated cells.
4. **Condition-specific genes:** Expressed in response to environmental signals (e.g., hormones).

Epigenetics

- Study of **heritable changes in gene expression** that **do not alter DNA sequence**.

Transcription Regulation in *E. coli*

Operon Concept (Jacob & Monod, Nobel Prize)

- **Operon:** Cluster of genes under one promoter.
- **Components:**
 - **Promoter:** RNA polymerase binding site
 - **Operator:** Regulatory protein binding site
 - **Structural genes:** Encode proteins
- **Regulation:** Proteins can **activate or repress transcription** by binding to the operator.

Types of Operons

1. **Inducible Operon (e.g., lac operon):**
 - Default: Repressor bound → transcription blocked.
 - Inducer removes repressor → transcription activated.
2. **Repressible Operon (e.g., trp operon):**
 - Default: Transcription occurs.
 - Corepressor (e.g., tryptophan) activates repressor → transcription blocked.
3. **Operons under Positive Control:**
 - Activator proteins (e.g., CRP–cAMP) enhance transcription.

Summary Table

Feature	Inducible	Repressible	Positive Control
Default state	OFF	ON	Low expression
Regulation	Repressor removed by inducer	Repressor activated by corepressor	Activator binds
Example	lac operon	trp operon	CRP–cAMP

Transcription Regulation in Prokaryotes (*E. coli*)

Types of Regulatory Systems

E. coli uses three major transcription regulatory systems:

1. **Inducible Operon (Repressor-regulated)**
 - Example: **lac operon**
 - Transcription is **off** until an inducer turns it **on**.
2. **Repressible Operon (Repressor-regulated)**
 - Example: **trp operon**
 - Transcription is **on** until a corepressor turns it **off**.
3. **Operon controlled by an Activator Protein**
 - Example: **CRP–cAMP system**

Negative Control of Transcription

Regulatory molecule (repressor) prevents transcription.

Example: trp Operon

- In high tryptophan conditions:
 - Tryptophan = **corepressor**
 - Activates repressor → binds operator → **transcription stops**
- **Repressible system** = normally on → turned off when product is abundant.

Positive Control of Transcription

Regulatory molecule enhances transcription

Example: CRP–cAMP Complex

- When glucose is low:
 - cAMP increases → binds CRP (activator)
 - Enhances binding of RNA polymerase → **transcription increases**
- Used mainly for sugar metabolism genes (lac operon activation requires this)

□ Transcription Regulation in Eukaryotes

The Eukaryotic Dilemma

Cells must:

- ✓ turn **on** the correct genes
- ✓ at the correct **time**
- ✓ in the correct **cells**

Chromatin Structure Matters

Genes are found in **chromatin**, which can exist as:

1. Euchromatin

- Loose
- Transcriptionally **active**

2. Heterochromatin

- Tightly packed
- Transcriptionally **inactive**

↻ DNA Organization

Most DNA is:

- Highly compact
- Wrapped around **positively charged histone proteins**
- Not accessible to RNA polymerase

↻ Chromatin Reorganization

Occurs when signals activate **transcription activators**

Steps:

1. **Activator binds a specific DNA sequence**
2. Works with other proteins to modify chromatin

Mechanisms of Chromatin Remodeling

1. **Histone Acetylation**
 - Adds acetyl groups ($-\text{COCH}_3$) to histones
 - Reduces positive charge
 - DNA becomes **less tightly bound**
→ genes become **more accessible**
2. **Chromatin Remodeling Complex (CRC)**
 - Actively shifts or loosens nucleosomes
 - Allows transcription factors + RNA polymerase to bind
3. **Activators can recruit CRC**
 - Enhances transcription efficiency

★ Key Differences: Prokaryotes vs Eukaryotes

Feature	Prokaryotes	Eukaryotes
Genome state	Not chromatin-wrapped	Chromatin-wrapped
Regulation type	Mainly operons	Chromatin + TF-based
Control level	Mainly transcription	Multi-level (epigenetic, post-transcriptional, etc.)
DNA accessibility	Easy	Restricted
Response	Fast	Complex + slower

1. Eukaryotic Transcription — Initiation

Promoter Elements

Eukaryotic promoter has two key sequences:

1. **Recognition sequence**
 - Recognized by **RNA polymerase** and transcription factors
2. **TATA box**
 - Site where DNA begins to **denature (unwind)**
 - Located **≈ 25 bp upstream** from the start site (in most promoters)

Role of Transcription Factors

- RNA polymerase II **cannot** initiate transcription alone.
- Requires **transcription factors (TFs)**.

Steps:

1. The protein **TFIID** binds first
2. TFIID includes **TBP (TATA-binding protein)**
3. Binding of TFIID causes:
 - Shape change in **TFIID**
 - Shape change in **DNA**
4. This exposes a surface for binding of additional TFs
5. A **transcription initiation complex** forms
6. **RNA polymerase II joins last**
7. Transcription begins

Enhancers

- Some genes have **enhancer sequences**
- Located far from promoter (sometimes **1000 bp away**)
- Bind **activator proteins** that are produced only in specific cell types
- Activation increases transcription efficiency

Simultaneous Regulation of Many Genes

- Different genes can share **common regulatory sequences**
- Same regulatory protein can bind to multiple promoters
→ Enables coordinated expression of widely separated genes

□ 2. Chromosomal Aberrations (Essentials of Genetics)

Definition

Chromosomal aberrations = Any **change in number or structure** of chromosomes.

Effects:

- Alters gene product amounts

- Can directly cause physical or developmental anomalies
- Can disrupt gene balance

Classification of Chromosome Aberrations

There are three broad groups:

1. **Numerical changes**
2. **Structural changes**
3. **Different cell lines (mosaicism)**

1) Numerical Changes

Includes:

- **Aneuploidy** (change in part of set)
- **Euploidy** (change in whole set)

Examples of numerical terms:

Term	Meaning
Monoploidy	One set (n)
Diploidy	Two sets (2n — normal)
Polyploidy	3+ sets (e.g., 3n, 4n, etc.)
Hypoploidy	Loss of chromosome(s)
Hyperploidy	Gain of chromosome(s)

Numerical changes are usually due to **de novo meiotic errors** (e.g., nondisjunction).

2) Structural Changes

Types:

- **Translocations**
- **Deletions**
- **Duplications**
- **Inversions**

These may occur:

- during meiosis (**de novo**) or
- be **inherited**

3) Different Cell Lines

- Called **mosaicism**
- Occurs **post-zygotically**
- Individual has two or more genetically different cell lines

☐ Karyotyping — Nomenclature

Normal Karyotype Example

For a male:

46, XY

Meaning:

- **46** = total chromosomes
- **XY** = sex chromosome constitution

Anomaly Description

Format:

Total chromosomes, Sex chromosomes, Abnormality

Example:

47, XY, +21

- Indicates **Down syndrome**
- Extra chromosome 21

☐ Essentials of Genetics

DEFINING CHROMOSOME REGIONS

Chromosome Regions Include

- **Arm**
- **Region**
- **Band**
- **Subband**

Example of notation:

17q11.2

Where:

- **17** = Chromosome number
- **q** = Long arm
- **1** = Region
- **1** = Band
- **.2** = Subband

Bands Within Bands

Staining techniques reveal:

- Large bands
- Subbands
- Sub-subbands

Helpful for:

- ✓ Gene localization
- ✓ Identifying chromosomal abnormalities

CHROMOSOME IDENTIFICATION

Done through **microscopic examination** → *Karyotype*

Chromosomes classified by:

1. **Size**
2. **Location of centromere**
3. **Banding pattern**

Banding patterns (G-bands, Q-bands, etc.) help identify structural defects.

□ COMMON NUMERICAL ABNORMALITIES

Numerical abnormalities arise due to:

- Change in chromosome **number**

Examples include:

Autosomal Trisomies

- **Down syndrome** → *Trisomy 21*

47, XX, +21 or 47, XY, +21
- **Edwards syndrome** → *Trisomy 18*

47, XX, +18 or 47, XY, +18

- **Patau syndrome** → *Trisomy 13*

47, XX, +13 or 47, XY, +13

Sex Chromosome Abnormalities

- **Turner syndrome** → *45, X*
- **Klinefelter syndrome** → *47, XXY*

Polyploidy

- **Triploidy** → *69 chromosomes*

□ DOWN SYNDROME (TRISOMY 21)

Key Points

- **47, +21**
- Risk increases with **maternal age**
- **3-4%** cases due to **Robertsonian translocation**
 - Not related to maternal age

Clinical Features of Down Syndrome

Head & Neck

- Brachycephaly
- Epicanthal folds
- Brushfield spots
- Flat nasal bridge
- Folded ears
- Open mouth with protruding tongue
- Short neck

Extremities

- Short broad hands
- Short curved (incurved) 5th finger
- Increased gap between 1st + 2nd toes
- Joint hyperflexibility

□ TURNER SYNDROME (45,X)

Key Facts

- Loss of one sex chromosome
- Affects only **females**
- Not typically **hereditary**
- No known environmental causes
- Women usually **infertile**

Clinical Features

- Short stature (143–145 cm)
- Loss of ovarian function
- Hormonal imbalance (thyroid/diabetes)
- Emotional stress sensitivity
- Kidney/heart/lung/gut diseases possible
- Bone disorders
- Learning difficulties (esp. mathematics)

□ TRIPLOIDY

- Total chromosomes = **69**
- Affects all chromosomes

□ NUMERICAL ABNORMALITIES SUMMARY

Caused by changes in:

→ Chromosome **number**

Often due to errors in:

- ✓ Meiosis (more common)
- ✓ Mitosis

NON-DISJUNCTION

Definition

Failure of homologous chromosomes or sister chromatids to separate during cell division.

Normally:

Disjunction separates chromosomes during meiosis/mitosis.

Failure → **Non-disjunction**

Types of Non-Disjunction

1. **Meiotic non-disjunction**
2. **Mitotic non-disjunction**

Meiotic Non-Disjunction

If during **meiosis I**:

- Homologous chromosomes fail to separate
→ All gametes are abnormal

If during **meiosis II**:

- Sister chromatids fail to separate

Results in:

- ✓ Trisomy
- ✓ Monosomy

Mitotic Non-Disjunction

Occurs after fertilization (post-zygotic)

Leads to:

- ✓ Mosaicism
- ✓ Different cell lines
- ✓ Trisomic + monosomic cells

Mechanisms:

- Spindle fiber failure
- Chromatid non-separation

Causes of Non-Disjunction

- Aging effect on primary oocyte
- Radiation exposure
- Delayed fertilization after ovulation

□ **Essentials of Genetics**

TRISOMY

Definition

- Trisomy is a type of **polysomy** where there are **three copies of a particular chromosome** instead of the normal two.
- Trisomy is also a form of **aneuploidy** (abnormal chromosome number).

Types of Trisomy (Based on Chromosome)

1. **Autosomal trisomy**
2. **Sex chromosome trisomy**

Example:

- **Trisomy 21 → Down syndrome**

Types of Trisomy (Based on Structure)

1. **Primary trisomy**
 - Entire extra chromosome is present.
2. **Partial trisomy**
 - Only part of a chromosome is duplicated.
3. **Secondary trisomy**
 - Extra chromosome contains **duplicated arms**.
4. **Tertiary trisomy**
 - Extra chromosome formed from arms of **two different chromosomes**.

Common Trisomies in Humans

- **Trisomy 21** (Down syndrome)
- **Trisomy 18** (Edward's syndrome)
- **Trisomy 13** (Patau's syndrome)
- **Sex chromosome trisomy**
 - Example: **Klinefelter syndrome (47,XXY)**

Trisomy 21 (Down Syndrome)

- Trisomy of chromosome 21
- **Incidence:** ~1 in 700 live births
- Associated with **increased maternal age**

Trisomy 18 (Edward's Syndrome)

- Caused by extra copy of chromosome 18

- Severe developmental abnormalities

Trisomy 13 (Patau's Syndrome)

- Caused by extra copy of chromosome 13
- Severe neurological and facial abnormalities

□ MONOSOMY

Definition

- Monosomy = **loss of a single chromosome**

Example:

- **Turner syndrome (45,X)** → monosomy of sex chromosome

Lethal Nature

- **Autosomal monosomy is lethal**
(because loss of gene dosage cannot be compensated)

Causes of Monosomy

1. **Non-disjunction**
 - One gamete receives **two homologous chromosomes**
 - Other gamete receives **none**
2. **Chromosome loss**
 - Chromosome may be lost as it moves toward pole during **anaphase**

Examples / Diseases due to Monosomy

- **Turner syndrome (45,X)**
- **Cri-du-chat syndrome**
 - Caused by partial deletion on chromosome 5 (5p deletion) but often classified with monosomic conditions because effect resembles loss of material

Other Reported Monosomies in Organisms

- Seen experimentally in **fruit fly (Drosophila)**

- Monosomy of chromosome **7** and **22** can occur in humans, often associated with cancer or severe developmental issues

✓ Quick Comparison Table

Feature	Trisomy	Monosomy
Definition	Extra chromosome (3 copies)	Missing chromosome (1 copy)
Type of condition	Aneuploidy	Aneuploidy
Frequency	More common	Rare
Autosomal viability	Some survive (21,18,13)	Mostly lethal
Sex chromosome viability	XXY, XYY possible	XO survives
Example	Down syndrome	Turner syndrome

□ Essentials of Genetics

POLYPLOIDY

Definition

- Polyploidy = presence of **multiples of the haploid chromosome number**
- Examples:
 - Triploid → **3x**
 - Tetraploid → **4x**
 - Hexaploid → **6x**

Viability

- **Triploidy or tetraploidy** in humans → **fetus does not survive**
- Very common and advantageous in **plants**

Types of Polyploidy

Two major types:

1. **Autopolyploidy**
2. **Allopolyploidy**

Examples of Polyploidy Levels

- **Triploid (3x)** → Watermelon
- **Tetraploid (4x)** → Cotton
- **Pentaploid (5x)** → Kenai Birch
- **Hexaploid (6x)** → Wheat

Causes of Polyploidy

Polyploidy can arise due to:

1. **Retention of polar body**
2. **Formation of diploid sperms**
3. **Dispermy** → ovum fertilized by **two sperms**

TRIPLOIDY

- Presence of **three sets of chromosomes**
- Common in plants, lethal in humans

POLYPLOIDY IN PLANTS

Plants frequently display polyploidy and often benefit from it.

Examples by Level

Triploid (3x) Plants

- Apple
- Banana
- Citrus
- Ginger
- Watermelon

Tetraploid (4x) Plants

- Cotton
- Potato
- Tobacco
- Peanut

Hexaploid (6x) Plants

- Oat
- Kiwifruit

Octaploid (8x) Plants

- Strawberry

Decaploid (10x) Plants

- Some sugar cane hybrids

□ ALLOPOLYPLOIDY

Two Types of Polyploidy

1. **Autopolyploidy**
2. **Allopolyploidy**

Autopolyploidy

- Chromosome sets derived from **one species**
- Example:
 - AAAA (tetraploid of same species)

Allopolyploidy

- Chromosome sets derived from **different species**
- Occurs through **interspecies crosses**

Key Features

- Interspecies hybrid offspring often **sterile**
 - Because homologous chromosome pairing fails during meiosis

Allodiploid Formation

- Cross between two species with different chromosome sets

Colchicine and Polyploidy

Use of Colchicine

- Used to promote formation of **polyploid and allopolyploid plants**
- Why?
 - Polyploid plants often show:
 - ✓ Larger fruits
 - ✓ Higher yield
 - ✓ Better vigor

Mechanism of Action

- Colchicine binds **tubulin**
- Disrupts microtubule formation
- Blocks chromosome segregation → induces **polyploidy**

✓ Summary

Concept	Description
Polyploidy	Multiples of haploid number
Autopolyploidy	Chromosomes from same species
Allopolyploidy	Chromosomes from different species
Common in	Plants
Lethal in humans	Triploidy, Tetraploidy
Induced by	Colchicine
Example crops	Wheat, cotton, strawberry

Euploidy & Aneuploidy — Summary

Aneuploidy

Definition:

A numerical chromosomal abnormality where the chromosome number changes by the **addition or deletion of individual chromosomes** or chromosome segments.

Types of Aneuploidy

- **Nullisomy:** loss of a pair ($2n - 2$)
- **Monosomy:** loss of one chromosome ($2n - 1$)
- **Trisomy:** gain of one chromosome ($2n + 1$)
- **Tetrasomy:** gain of two homologous chromosomes ($2n + 2$)

Euploidy

Definition:

A condition in which an organism gains **one or more complete sets of chromosomes**.

Types of Euploidy

- **Triploid (3n)**
- **Tetraploid (4n)**
- **Pentaploid (5n)**
(Polyploids are more common in plants)

Comparison

Feature	Aneuploidy	Euploidy
Change	Individual chromosomes	Whole sets of chromosomes
Amount	Unbalanced change	Balanced change in sets
Example	Trisomy 21 (Down syndrome)	Triploidy
Origin	Nondisjunction	Genome duplication

Structural Chromosomal Abnormalities — Summary

Definition

Structural abnormalities occur due to **chromosome breakage and rejoining** in abnormal patterns, causing rearrangements.

Based on Genetic Material

1. **Balanced:**
 - No genetic material lost or gained
 - Usually does not affect phenotype (unless break disrupts a gene)
2. **Unbalanced:**
 - Gain or loss of genetic material
 - Usually affects phenotype

Types of Structural Abnormalities

- **Translocations**
- **Deletions**
- **Insertions**
- **Inversions**
- **Ring Chromosomes**
- **Isochromosomes**

- ✓ **Translocations**
- ✓ **Chromosomal Deletions**
- ✓ **Chromosomal Duplications**

I'll keep everything accurate and complete, but easier to revise.

Essentials of Genetics — Structural Abnormalities

1. Translocations

Definition

A **translocation** is the **transfer of genetic material from one chromosome to another**.

Types of Translocations

1. **Reciprocal translocation**
2. **Robertsonian translocation**

Balanced vs. Unbalanced Translocation

- **Balanced:**
 - No gain or loss of genetic material
 - Usually symptomless / normal phenotype
- **Unbalanced:**
 - Gain or loss of genetic material
 - Often leads to abnormalities, disease, infertility, or miscarriage

a. Reciprocal Translocations

- Exchange of genetic material between **two non-homologous chromosomes**
- Usually **balanced**
- Occurs in roughly **1 in 600 live births**

Mechanism:

Chromosomes break → exchange segments → rejoin in new configuration

b. Robertsonian Translocations

- Occur between **two acrocentric chromosomes**
- Breaks occur at the **short arms**
- Small fragments are **lost**
- Larger fragments **fuse at the centromere**

Common Human Example:

- **Chromosome 13 & 14**
- Frequency $\approx$ **0.97 per 1000 newborns**
- Can lead to hereditary **Down syndrome**

Human Disorders Associated with Translocations

- Cancer
- Infertility
- Down syndrome
- Leukemia
- XX male syndrome

2. Chromosomal Deletions

Definition

A deletion involves the **loss of one or more genes** or a portion of a chromosome.

- Large deletions often **incompatible with life**

Causes

- **10–15%** arise from balanced parental translocations
- **85–90%** are true (de novo) deletions

Types of Deletions

1. **Terminal deletion**
 - Loss from one end of the chromosome
 - Example: **Cri-du-chat syndrome**
2. **Interstitial deletion**
 - Two breaks → middle fragment lost
 - Example: deletions at chromosome 16
3. **Microdeletions**
 - Very small, detectable only by molecular techniques (FISH, microarrays, etc.)

Phenotypic Consequences Depend On:

- **Size** of deletion
- **Genes** lost
- Usually **detrimental**

3. Chromosomal Duplications

Definition

A duplication involves the **addition of genetic material**, where a gene or region is present **more than once**.

Evolutionary Significance

Duplications are a major source of **new genetic material** for molecular evolution (can give rise to new gene functions).

Causes of Duplication

Duplication can arise due to errors in:

- DNA replication
- DNA repair mechanisms

Common Mechanism:

Ectopic homologous recombination (unequal crossing-over)

→ Occurs during meiosis between **misaligned homologous chromosomes**

→ Produces one chromosome with **duplication** and the other with a **deletion**

Chromosomal Duplications (continued)

Other Causes of Duplication

In addition to unequal crossing-over, duplications may arise from:

- **Retrotransposition events**
- **Aneuploidy**
- **Polyploidy**
- **Replication slippage**

Replication Slippage

- A DNA replication error that generates **short sequence duplications**
- Occurs when **DNA polymerase slips** while copying repetitive sequences
- Often results in **tandem repeats**

Types of Duplications

1. **Dispersed duplications**
 - Duplicate segments located **far apart** on the same or different chromosomes
2. **Tandem duplications**
 - Copies are located **adjacent** to each other

Fate of Duplicated Genes

Duplicated copies may:

- Acquire **modified/new functions**
- Become **pseudogenes** (nonfunctional) by mutation
- Play a role in **genetic novelty and evolution**

Chromosomal Inversions

Definition

An inversion occurs when:

- A chromosome undergoes **two breaks**
- The segment between the breaks **reverses orientation**
- The region is reinserted into the chromosome

Types of Inversions

1. **Pericentric inversion**
 - **Centromere included** in the inverted segment
2. **Paracentric inversion**
 - Inversion occurs on **one arm** and **does not include** the centromere

Genetic Effects of Inversions

- **No loss of genetic material**
- Often **no phenotypic consequences**
- Mutation at the **breakpoints** can disrupt genes or regulatory regions
- Can alter **gene expression** by repositioning genes

Inversion Heterozygotes

- Have **one normal** and **one inverted** chromosome
- Usually phenotypically normal
- Problems may arise during **meiosis**, affecting fertility

Examples in Humans

- ~ **2% of individuals** carry detectable inversions
- **inv(9)(p12q13)** is the most common
- Usually benign but associated with **subfertility or miscarriage risk**

Cell Biology — Cytoskeleton

The cytoskeleton maintains **cell structure, shape, and movement**.

Components

1. **Actin filaments (microfilaments)**
2. **Intermediate filaments (IF)**
3. **Microtubules**

1. Actin Filaments

Functions:

- Stabilize cell shape (cortical actin)
- Local cellular movements
- **Muscle contraction**
- **Cytokinesis** (contractile ring)
- **Cytoplasmic streaming**

2. Intermediate Filaments (IF)

Functions:

- Provide **mechanical strength**
- Help maintain **tissue rigidity**
- Anchor organelles
- Form **desmosomes**
- **Lamins** support nuclear envelope
- Resist **tension**

3. Microtubules

Functions:

- Provide **rigid internal support**
- Resist **compression**
- Serve as **tracks for motor proteins**
- Key component of **mitotic spindle**

Structure:

- Made of **α -tubulin + β -tubulin dimers**
- Have **plus (+)** and **minus (-)** ends

Centrosome:

- Main **microtubule organizing center (MTOC)** in animal cells

Movement Across Cell Membranes — Simple Diffusion

Definition

Simple diffusion is the:

Movement of molecules from **high to low concentration** due to random motion

Energy Requirement

- **Passive transport**
- Requires **no ATP**
- Driven by **concentration gradient**

Role in Cells

Simple diffusion is important for **short-distance movement** of molecules:

- **O₂** diffusion from lungs → blood
- **CO₂** diffusion from tissues → extracellular fluid

Movement Across Cell Membranes — Osmosis

Contents

1. Definition
2. Osmotic Pressure
3. Hyperosmotic
4. Isosmotic
5. Hypoosmotic
6. Role of Osmotic Pressure in Cells

1. Osmosis — Definition

Osmosis is the **movement of water molecules** across a **selectively permeable membrane** from a region of **higher water concentration (low solute)** to a region of **lower water concentration (high solute)**.

Key point: Solute pulls water.

2. Osmotic Pressure

- **Osmotic pressure** is the **minimum pressure** required to prevent water from flowing into a solution across a semipermeable membrane.
- Higher solute concentration → **higher osmotic pressure**

3. Hyperosmotic

A solution is **hyperosmotic** when it has **higher solute concentration** (and lower water concentration) compared to its surroundings.

Example:

- If a **cell is placed in pure water**, the **cell is hyperosmotic** relative to the surrounding water.

Effect on cell:

- Water moves **into the cell**, causing it to swell.

4. Isosmotic

A solution is **isosmotic** when it has the **same solute concentration** as its surroundings.

Example:

- **Cell in physiological saline (0.9% NaCl)** is typically **isosmotic** to its environment.

Effect:

- **No net movement of water** → cell volume remains stable.

5. Hypoosmotic

A solution is **hypoosmotic** when it has **lower solute concentration** (higher water concentration) than its surroundings.

Example:

- Cell in **concentrated salt solution** → the **cell is hypoosmotic** relative to the environment.

Effect:

- Water moves **out of the cell**, causing it to shrink (crenation in animals; plasmolysis in plants).

6. Role of Osmotic Pressure in Cells

Osmosis is essential for:

- Maintaining **ionic balance**
- Maintaining **osmotic equilibrium**
- Uptake of **minerals and nutrients**
- Proper **cell volume regulation**

Movement Across Cell Membranes

A. Facilitated Diffusion

Definition

Facilitated diffusion is the **movement of molecules from high concentration to low concentration** across a membrane **with the help of proteins**.

- Passive transport (no energy required)
- Driven by concentration gradient

Membrane Proteins Involved

Two major protein types assist transport:

1. **Channel proteins**
 - Form pores for ions or water
 - e.g., Na⁺ channels, K⁺ channels, aquaporins
2. **Carrier proteins**
 - Bind to specific molecules (e.g., glucose) and undergo conformational change

Energy Requirement

- **No ATP required** (passive)
- Only down concentration gradient

Role in Cells

Facilitated diffusion is used when:

- Molecules are **too large** for simple diffusion (e.g., glucose)
- Molecules are **charged or polar** and cannot cross lipid bilayer

Examples:

- Glucose transport (GLUT transporters)
- Na⁺ and K⁺ ion channels

B. Filtration

Definition

Filtration is the process where **hydrostatic pressure forces small molecules** across a selectively permeable membrane.

Hydrostatic Pressure

- Pressure exerted by a fluid
- Acts in all directions
- Increases with depth

Filtration in Blood Capillaries

- Blood pressure pushes **water and dissolved molecules** through capillary walls
- Proteins and large molecules cannot pass through small pores

Filtration in Kidneys

- First step of urine formation = **pressure filtration**
- Blood pressure forces water and solutes into kidney tubules (nephrons)

C. Active Transport

Definition

Active transport is the movement of molecules **against concentration gradient** (from low to high concentration) across a membrane.

Energy Requirement

- Requires **ATP**
- Not spontaneous

Role of Membrane Proteins

- Uses **carrier proteins called pumps**
- Transport is **active**, not passive

Examples

Na⁺/K⁺ Pump

- Maintains **high K⁺ inside** and **high Na⁺ outside** the cell
- Essential for **nerve impulse transmission**

Ca²⁺ Pump

- Maintains low Ca²⁺ concentration in cytosol

D. Endocytosis

Definition

Endocytosis is **bulk transport of material into the cell** by vesicle formation from the plasma membrane.

Three major types:

1. Pinocytosis

- “Cell drinking”
- Uptake of **extracellular fluid and dissolved solutes**
- Forms small vesicles

2. Phagocytosis

- “Cell eating”
- Uptake of **solid particles** (e.g., microbes, food particles)
- Vesicle fuses with **lysosome** → **phagolysosome** for digestion

3. Receptor-Mediated Endocytosis

- Involves **specific receptors** on plasma membrane
- Highly selective
- Example: **cholesterol transport (LDL receptors)**

E. Exocytosis

Definition

Exocytosis is the process by which **vesicles fuse with plasma membrane and release contents outside the cell.**

- Reverse of endocytosis

Functions

1. Secretion

- Export of proteins and enzymes synthesized in the cell
- Golgi apparatus packages them into vesicles

2. Waste Removal

- In unicellular organisms (e.g., amoeba, paramecium), undigested wastes are expelled via exocytosis

3. Membrane Replacement

- Adds new membrane material to plasma membrane
- Balances membrane loss during endocytosis

CYTOKINESIS

Definition

Cytokinesis is the **process by which a cell divides its cytoplasm to produce two daughter cells.**

- It is the **final stage of the cell cycle**, completing **cell division** after mitosis or meiosis.

Key Points

- Cytokinesis usually **follows every mitosis**.
- Some cells (e.g., **early Drosophila embryos, certain hepatocytes, and cardiac muscle cells**) can undergo **mitosis without cytokinesis**, resulting in **multinucleated cells**.

Cytokinesis in Animal Cells

Cleavage Furrow

- A **cleavage furrow** appears at the cell surface.
- It **deepens and spreads**, eventually splitting the cell into **two daughter cells**.

Contractile Ring

- Made of **actin and myosin II filaments**.
- Present in **interphase cells** as a cortical network under the plasma membrane.
- Disassembles during **mitosis** and later reforms to drive cytokinesis.

Role of Microtubules

- **Mitotic spindle microtubules** determine the **plane of cell division**.
- Ensures division occurs at the **correct time and place**.

Models of Contractile Ring Positioning

- Signals from **anaphase spindle microtubules** guide the **formation and placement of the contractile ring**.

Cytokinesis in Plant Cells

Challenges

- Plant cells have a **rigid cell wall**, so they **cannot form a cleavage furrow** like animal cells.

Cell Plate Formation

- A **cell plate** forms across the middle of the cell during **telophase**.
- Gradually develops into **two separate cell walls**, splitting the cell into **two daughter cells**.

Phragmoplast

- Guides **cell plate formation** in higher plants.
- Contains **microtubules from the mitotic spindle**.

- Assembles in **late anaphase**, directing the formation of the **new cell walls**.

Summary

Feature	Animal Cells	Plant Cells
Cytoplasm division	Cleavage furrow	Cell plate
Contractile structure	Actin-myosin ring	Phragmoplast (microtubules)
Mechanism	Squeezing	Vesicle fusion → new cell wall
Timing	After mitosis	During late anaphase → telophase

CELL CYCLE CONTROL SYSTEM

Definition

The **cell cycle control system** ensures that **all events in the cell cycle occur in the correct order**.

- Coordinates **DNA replication, mitosis, and cytokinesis**.
- Operates like a **timer and switch**, with ON/OFF signals for **irreversible transitions**.

Irreversible Points in the Cell Cycle

1. **Replication of genetic material** (S phase)
2. **Separation of sister chromatids** (anaphase)

Checkpoints

- Cell cycle is monitored by **STOP and GO signals** at key points.
- Ensure **cellular processes are completed correctly** before moving forward.

Major checkpoints:

Checkpoint	Function
START (G1/S)	Cell commits to entering the cycle; decides whether to divide
G2/M	Triggers early mitotic events; checks DNA replication
Metaphase → Anaphase	Ensures proper spindle attachment and triggers sister chromatid separation

Gap Phases

- **G1 phase:** Gap between M and S; cell grows and prepares for DNA replication.
- **G2 phase:** Gap between S and M; cell checks DNA for proper replication and repairs damage.

Cell Cycle Phases:

1. **G1** – growth and preparation
2. **S** – DNA synthesis
3. **G2** – preparation for mitosis
4. **M** – mitosis and cytokinesis

Cyclins and Cdks

Cyclin-dependent kinases (Cdks)

- **Protein kinases** that trigger cell-cycle events.
- **Inactive alone**; activated when bound to **cyclin**.

Cyclins

- Proteins that **cycle in synthesis and degradation** each cell cycle.
- Control the **timing of Cdk activation**.

Cyclin–Cdk Complexes

- Key regulators of **cell-cycle transitions**.
- **Cyclic Cdk activity** ensures orderly progression.

Four classes of cyclins:

Cyclin Type	Phase	Function
G1-cyclins	Late G1	Promote passage through Start/Restriction point
G1/S-cyclins	Late G1 → S	Activate Cdks to commit to DNA replication
S-cyclins	S phase	Stimulate chromosome duplication
M-cyclins	M phase	Promote mitotic events

Key Features of the Control System

- Functions as a **binary ON/OFF switch** for critical events
- **Highly adaptable**; can be modified for different cell types
- Ensures **accuracy and irreversibility** of cell cycle events

✓ Quick Tip:

- Think of **G1/S = decision to replicate**, **S = DNA duplication**, **G2/M = preparation for division**, **M = mitosis & cytokinesis**.
- Cyclins are like “keys” that **turn Cdks on** to unlock each phase.

CONTROL OF CELL DIVISION

Purpose

- Ensures **cells divide at the right time** and maintain proper size.
- Each daughter cell is an **exact replica** of the parent cell with the same genetic information.

Determinants of Organism/Organ Size

1. **Cell number** – controlled by **cell division and cell death**.
2. **Cell size** – growth before division.
3. **Total cell mass** – combination of number and size of cells.

Fundamental processes regulating size:

- **Cell growth**
- **Cell division**
- **Cell survival**

Extracellular Regulation

Cells respond to **extracellular molecules** that control growth, division, and survival:

Three major classes of extracellular signals:

Signal Type	Function
Mitogens	Stimulate cell division by activating G1/S-Cdk and mitosis. Reveal intracellular negative controls.
Growth factors	Promote cell growth by increasing synthesis of proteins/macromolecules and inhibiting their degradation.
Survival factors	Promote cell survival by inhibiting apoptosis (programmed cell death).

Key Points in Cell Cycle Regulation

- **G1 phase** is critical for controlling cell division.
- **Late G1 checkpoint** (just before S phase) ensures DNA replication only occurs when conditions are suitable.
- **Nutrients and growth factors**: their absence inhibits division.

Multicellular vs. Unicellular Organisms

Organism Type	Division Behavior
Unicellular	Divide as fast as possible to reproduce.
Multicellular	Divide only when more cells are needed; require stimulatory signals from neighbors .

Examples of Mitogens

- **PDGF (Platelet-Derived Growth Factor)** – stimulates cell proliferation.
- **EGF (Epidermal Growth Factor)** – acts on epidermal and many other cell types.
- There are **50+ known animal mitogens**.

Summary

- **Cell division is tightly controlled** by internal checkpoints and extracellular signals.
- **Mitogens, growth factors, and survival factors** coordinate **cell proliferation, growth, and survival**.
- Proper regulation ensures **organism size, tissue function, and prevention of uncontrolled growth (cancer)**.

DNA Damage Blocks Cell Division

How DNA Damage Occurs

DNA can be damaged by:

1. **Spontaneous chemical reactions** in DNA
2. **Errors during DNA replication**
3. **Exposure to radiation or certain chemicals**

Detection and Signaling: ATM & ATR

- DNA damage **activates protein kinases**:
 - **ATM** (Ataxia Telangiectasia Mutated)
 - **ATR** (ATM and Rad3-related)
- These kinases **initiate signaling pathways** that halt the cell cycle to allow repair.

Cell-Cycle Arrest Mechanisms

1. **p53 Pathway**
 - DNA damage stabilizes **p53**, a key regulatory protein.
 - Normally, p53 is degraded by **Mdm2** (a ubiquitin ligase).
 - Phosphorylation of p53 **reduces Mdm2 binding**, increasing p53 levels.
 - p53 activates transcription of **p21** (a CKI protein).
 - **p21 binds G1/S-Cdk and S-Cdk complexes**, blocking progression from G1 to S phase.
2. **Chk1 & Chk2 Kinases**
 - These phosphorylate **Cdc25 phosphatases**, inhibiting their activity.
 - This **blocks entry into mitosis**, preventing cell division until damage is repaired.

Effect of DNA Damage

- DNA damage **halts cell cycle progression** at checkpoints (G1 or G2/M).
- Prevents replication of damaged DNA.
- Ensures genomic stability.

Cancer-Promoting Mutations

- Low-level DNA damage is normal, but **accumulated damage increases cancer risk**.
- Mutations in **p53** occur in **~50% of human cancers**, leading to loss of cell cycle arrest and uncontrolled growth.

Key Points

Component	Function in DNA Damage Response
ATM / ATR	Detect DNA damage and activate checkpoint signaling
p53	Transcription factor; induces p21 to block G1/S progression
Mdm2	Ubiquitin ligase; degrades p53 under normal conditions
p21	Cyclin-dependent kinase inhibitor (CKI); halts cell cycle
Chk1 / Chk2	Phosphorylate Cdc25, blocking mitotic entry

DNA Mutations

Definition

- A **DNA mutation** is a **permanent, transmissible change** in the genetic material (DNA or RNA).
- It is a **permanent change in the nucleotide sequence** of an organism's genome.

Causes of DNA Damage / Mutation

1. Endogenous Agents (internal)

- Produced naturally **inside cells** during metabolic processes.
- Examples: reactive oxygen species (ROS), replication errors, spontaneous chemical changes.

2. Exogenous Agents (external)

- Originating from the **environment**.
- Examples: UV radiation, X-rays, chemical mutagens, toxins.

Significance of DNA Stability

- **Maintains genetic fidelity**: Ensures accurate transmission of information from one generation to the next.

- **Maintains gene function:** Correct functioning of thousands of genes depends on DNA integrity.

Significance of DNA Mutations

- **Source of genetic variation:** Drives evolution.
- Without mutations, **new species could not arise**.
- **Balance is key:** Life and biodiversity depend on a balance between DNA mutation and DNA repair.

Key Points

Aspect	Role / Importance
DNA Mutation	Permanent change in nucleotide sequence
Endogenous Damage	From normal metabolism (e.g., ROS)
Exogenous Damage	From environment (e.g., UV, chemicals)
DNA Stability	Ensures fidelity of genetic info and proper gene function
DNA Variation	Needed for evolution and species diversity
Balance	Mutation vs. repair maintains life and biodiversity

Nature of DNA Mutations

1. Scale of Mutations

- Mutations can be **simple** (single base changes) or **complex**, affecting thousands of nucleotides.

2. Point Mutations

- **Point mutations** alter a single nucleotide. Types include:
 1. **Transitions:** pyrimidine ↔ pyrimidine (T ↔ C) or purine ↔ purine (A ↔ G)
 2. **Transversions:** pyrimidine ↔ purine (T ↔ A or G, C ↔ A or G)
- **Insertions & deletions** of a few nucleotides are also point mutations.

3. Large-Scale / Complex Mutations

- Can involve **extensive insertions, deletions, or rearrangements** of DNA.
- Example: **Transposon insertion** introduces thousands of nucleotides into coding or regulatory sequences.
- **Chromosomal-level mutations:** Structural changes in chromosomes (appearance, number, or arrangement).

Causes of Mutations

1. Replication Errors

- DNA polymerase has a **proofreading mechanism** to remove misincorporated nucleotides.
- Some errors escape proofreading → mismatch remains → becomes **permanent after next replication**.

Mechanism:

1. Mismatch occurs during replication.
2. If uncorrected, it directs complementary base incorporation in the next replication.
3. Result → permanent mutation in DNA sequence.

2. Radiation Damage

- **High-energy radiations** can damage DNA:

Type	Wavelength	Notes
UV light	100–400 nm	Non-ionizing; mostly UV-A reaches earth, little damage. UV-B (~10% reaches earth) causes most skin DNA damage. UV-C (100–295 nm) highly energetic, mostly blocked by ozone.
X-rays	0.01–100 nm	Ionizing; can penetrate tissues and cause DNA breaks.
Gamma rays	<0.01 nm	Ionizing; highly penetrative, causes DNA breaks.

- **UV Light Bands:**
 - **UV-A (321–400 nm)**: least energetic, minor DNA damage.
 - **UV-B (296–320 nm)**: main cause of DNA damage in skin.
 - **UV-C (100–295 nm)**: highly energetic, absorbed by DNA (260 nm peak), mostly blocked by ozone.

Note: UV-C is used in labs (germicidal lamps) for sterilization.

Key Points

Category	Description
Simple mutation	Single base change (transition/transversion)
Point mutation	Small insertions or deletions
Complex mutation	Large insertions, deletions, or rearrangements
Chromosomal mutation	Structural changes in chromosomes
Replication error	Misincorporated nucleotides → permanent mutation if uncorrected
Radiation damage	UV, X-rays, gamma rays → DNA damage (strand breaks, base modification)

DNA Damage by X-rays & Gamma Rays (Ionizing Radiation)

1. Why X-rays & Gamma Rays are Hazardous

- These are **ionizing radiations**.
- They can cause **double-strand breaks (DSBs)** in DNA, which are **difficult to repair**.
- DSBs are a major cause of **cell death** and **chromosomal aberrations** (deletions, duplications, inversions, translocations).

2. Mechanisms of DNA Damage

DNA damage occurs **directly** or **indirectly**:

a. Direct Damage

- Radiation is **absorbed by DNA** or water bound to DNA.
- Leads to **strand breaks or base modifications**.

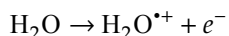
b. Indirect Damage

- Radiation is **absorbed by water molecules** (majority effect, ~65% of total DNA damage).
- Water molecules generate **reactive species** (free radicals) that attack DNA.

3. Reactive Species Produced

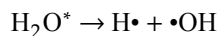
Two primary reactions occur when photons interact with water:

Reaction 1: Ionization of Water



- $\text{H}_2\text{O}^{*+} \rightarrow \text{H}^{+} + \bullet\text{OH}$ (hydroxyl radical)
- $e^{-} + \text{O}_2 \rightarrow \bullet\text{O}_2^{-}$ (superoxide radical)

Reaction 2: Excitation of Water



- Produces **H• (hydrogen radical)** and **•OH (hydroxyl radical)**

Overall reactive species:

- **•OH (hydroxyl radical)**
- **H• (hydrogen radical)**
- **•O₂⁻ (superoxide radical)**

4. DNA Lesions

- **Isolated lesions:** Single break or modification.
- **Clustered lesions:** Multiple lesions within a few helical turns; more harmful.
- **Double-strand breaks (DSBs):** Most lethal; primary cause of ionizing radiation's cytotoxicity.
- Leads to **chromosomal aberrations:** deletions, duplications, inversions, translocations.

5. Key Points

Feature	Description
Radiation type	X-rays, gamma rays (ionizing)
Main damage	Double-strand breaks
Mechanism	Direct (DNA absorption) / Indirect (water radicals)
Reactive species	$\bullet\text{OH}$, $\text{H}\bullet$, $\bullet\text{O}_2^-$
Consequences	Cell death, chromosomal aberrations, mutations

Tip for remembering:

“Ionizing radiation hits water → radicals form → radicals attack DNA → double-strand breaks → mutations or cell death.”

Mismatch Repair (MMR) System

Purpose

- Corrects rare **base pair mismatches** and **short insertions/deletions** that occur during DNA replication.
- Ensures high **replication fidelity** and prevents mutations.

1. Errors During Replication

- DNA polymerases make **~1 mispaired nucleotide per 10^5 bases**.
- Proofreading by $3' \rightarrow 5'$ exonuclease increases accuracy **100-fold**.
- Even a low error frequency (~ 1 per 10^7) can cause mutations over many cell divisions.
- Short insertions/deletions arise due to misalignment of **repeated sequences** during replication.

2. Consequences of Nonfunctional MMR

- Cells lacking functional MMR have **high mutation rates**.
- Increased risk of **genomic instability** and disease (e.g., cancer).

3. MMR in *E. coli*

- *E. coli* can distinguish **newly synthesized strand** from parental strand:
 - Only parental strand is **methylated** at GATC sequences.
 - Newly synthesized strand is **temporarily unmethylated**, allowing repair machinery to identify it.
- **Deoxyadenosine methylase** transfers methyl groups to GATC sites **after ~2 min**, creating a window for mismatch repair.

4. Key Proteins in *E. coli* MMR

Protein	Function
MutS	Recognizes and binds mismatch; homodimer or homotetramer
MutL	Recruited by MutS in ATP-dependent manner; coordinates repair
MutH	Endonuclease; cuts unmethylated GATC site near mismatch

Additional enzymes:

- DNA helicase II (UvrD) – unwinds DNA
- Single-strand binding protein (SSB) – stabilizes ssDNA
- Exonucleases – remove damaged strand:
 - 5'→3': ExoVII, RecJ
 - 3'→5': ExoI, ExoVII, ExoX
- DNA polymerase III holoenzyme – fills the gap
- DNA ligase – seals the nick
- Deoxyadenosine methylase – methylates repaired GATC

5. MMR Repair Process

1. **Mismatch recognition:** MutS binds the mismatch.
2. **Recruitment:** MutS recruits MutL (ATP-dependent).
3. **Strand incision:** MutS•MutL complex activates MutH → cuts nearest unmethylated GATC site.
4. **Unwinding & stabilization:** Helicase II unwinds DNA; SSB binds.
5. **Excision:**
 - If incision is 5' to mismatch → ExoVII or RecJ excises 5'→3'.
 - If incision is 3' to mismatch → ExoI, ExoVII, or ExoX excises 3'→5'.
6. **Resynthesis:** DNA polymerase III fills in new DNA.
7. **Sealing & methylation:** DNA ligase seals nick; methylase methylates GATC site.

6. Homologs in Other Organisms

- **MutS and MutL** homologs are present in **all organisms** with MMR.
- **MutH** is mostly restricted to ***E. coli* and some gram-negative bacteria**.

Key Concept for Exams:

"MutS detects → MutL coordinates → MutH nicks → exonucleases remove → DNA pol III fills → ligase seals → methylation restores parental marks."

Mismatch Repair in Eukaryotes

Key Differences from E. coli

- Eukaryotes **lack MutH**.
- They have **MutS and MutL homologs**:
 - MutS homologs: **MSH2, MSH3, MSH6**
 - MutL homologs: **MLH1, PMS2**

1. MutS Homologs

Heterodimer	Components	Function
MutSα	MSH2 + MSH6	Recognizes single base mismatches and small insertions/deletions
MutSβ	MSH2 + MSH3	Recognizes larger insertion/deletion loops

- Structure of MutS α/β is **similar to bacterial MutS**.

2. MutL Homologs

Heterodimer	Components	Function
MutLα	MLH1 + PMS2	Functions as latent endonuclease , activated by MutS α to nick DNA at mismatch sites

- PMS2 contains the **endonuclease motif**; MLH1 does not.
- MutL α **targets the discontinuous (newly synthesized) DNA strand**.

3. Human MMR Repair Process

Proteins involved:

- MutS α (mismatch recognition)
- MutL α (endonuclease activity)
- ExoI (5'→3' exonuclease)
- RPA (stabilizes single-stranded DNA)
- PCNA & RFC (clamp and loader proteins)
- DNA polymerase δ (fills in DNA)
- DNA ligase (seals nicks)

Step-by-Step Repair:

1. **Mismatch detection:** MutS α binds the mismatch.
2. **Activation of MutL α :** MutS α , PCNA, and RFC activate MutL α endonuclease.

3. **Strand incision:** MutL α nicks the newly synthesized (discontinuous) strand on **either 5' or 3' side** of the mismatch.
4. **Excision:**
 - 5'→3' exonuclease (ExoI) removes the mismatch.
 - RPA helps ExoI remove mismatch efficiently.
5. **Termination:** Once mismatch is removed, RPA inhibits ExoI.
6. **Resynthesis & ligation:** DNA polymerase δ fills in the gap; DNA ligase seals the nick.

4. Bidirectional Repair

- If the nick is on the **5' side** of the mismatch: MutS α + ExoI + RPA are sufficient.
- If the nick is on the **3' side**: MutL α , PCNA, RFC are required to create a new 5' entry point for ExoI.

5. Clinical Significance

- **Defective MutS α or MutL α → non-functional MMR**
- Leads to **hereditary nonpolyposis colon cancer (HNPCC / Lynch syndrome)**.
- HNPCC is **autosomal dominant**, characterized by early-onset colorectal cancer.

Key Concept for Exams:

"Eukaryotic MMR is MutS α /MutS β for recognition → MutL α for strand incision → ExoI excises → DNA pol δ fills → ligase seals."

DNA Repair Genes: TP53 / p53

Key Functions of p53

p53 is a **tumor suppressor gene** and “guardian of the genome.” Its main functions include:

1. **DNA Damage Response**
 - Activates DNA repair proteins when DNA is damaged.
 - Allows time for repair by **arresting the cell cycle at G1/S checkpoint**.
2. **Cell Cycle Arrest**
 - p53 acts as a transcription factor → induces **p21**, a cyclin-dependent kinase inhibitor (CKI).
 - **p21 inhibits Cdk1, Cdk2, Cdk4, Cdk6-cyclin complexes**, causing cell cycle arrest at **G1 and G2**.
3. **Apoptosis**
 - If DNA damage is irreparable, p53 can trigger **programmed cell death**.
4. **Anti-proliferative Effects**
 - Prevents abnormal cell division by inhibiting cyclin-Cdk complexes.

Regulation of p53

- **Normal cells:** p53 is unstable and degraded via Mdm2-mediated ubiquitination.

- **DNA damage:** Protein kinases phosphorylate p53 → prevents Mdm2 binding → p53 **accumulates**.
- **High Myc levels:** Activate **p19ARF**, which inhibits Mdm2 → increased p53 → cell-cycle arrest or apoptosis.

Clinical Significance

- **Li-Fraumeni Syndrome:** One p53 allele mutated → ~25x increased risk of cancer by age 50.
- **Viral interference:** Viruses (HPV, SV40) produce proteins (E6, E7, large T antigen) that **sequester p53 and Rb**, disabling the cell's anti-proliferative brakes.

Apoptosis Genes

Apoptosis Overview

- Essential for **cell turnover and tissue sculpting** (e.g., digits in developing embryos).
- **Balances cell division** in adults.

Caspase Cascade

- Apoptosis is executed by **caspases**, which are **cysteine proteases**:
 1. Synthesized as **inactive procaspases** with prodomain.
 2. Activated by **cleavage at specific sites** → forms **active caspases** (large + small subunits).
 3. **Amplification cascade**:
 - One active caspase X activates multiple caspase Ys → even more caspase Z molecules.
 4. Caspases cleave:
 - **Cytosolic proteins**
 - **Nuclear lamins** → nuclear breakdown
- Leads to **programmed cell death**.

Mechanisms of Apoptosis

1. **Intrinsic (mitochondrial) pathway:** triggered by internal signals (DNA damage, growth factor deprivation).
2. **Extrinsic (death receptor) pathway:** triggered by extracellular signals binding to death receptors.

Key Concept for Exams:

“p53 senses DNA damage → arrests cell cycle via p21 → repairs DNA or triggers apoptosis → caspases execute cell death.”

Genetics: Introduction & Historical Background

- **Character:** Observable feature (e.g., flower color).
- **Trait:** Specific form of a character (e.g., white flowers).
- **Heritable trait:** Passed from parent to offspring.

Mendelian Genetics

Monohybrid Cross

- **Parental generation (P):** True-breeding (e.g., SS × ss).
- **First filial generation (F1):** All heterozygous (Ss), dominant trait expressed.
- **Second filial generation (F2):** Traits reappear in 3:1 ratio (dominant:recessive).

Key Conclusions (Mendel)

1. Each plant has **two units of inheritance** (genes) for each character, one from each parent.
2. During gamete formation, only **one allele per gene** is passed to a gamete.
3. Mendel's **unit of inheritance** = **gene**, with **alleles** = **different forms** (S & s).
4. **Homozygous:** Two identical alleles (SS or ss).
5. **Heterozygous:** Two different alleles (Ss).
6. **Phenotype:** Physical appearance; result of **genotype**.

Punnett Square

- Diagrammatic method to predict offspring genotypes and phenotypes.

Mendel's Laws of Inheritance

First Law: Law of Segregation

- Alleles **separate during gamete formation**, so each gamete receives only one allele.

Second Law: Law of Independent Assortment

- Alleles of **different genes assort independently** during gamete formation.
- Segregation of gene A is independent of gene B.

Probabilities

- Certain event: 1
- Impossible event: 0
- Otherwise: between 0 and 1

Dihybrid Cross

- Cross involving **two characters**.
- Recombinant phenotypes appear according to independent assortment.

Aberrant Phenomena

1. **Incomplete Dominance:** Heterozygote shows **intermediate phenotype** (e.g., red × white → pink).
2. **Codominance:** Both alleles **express equally** in heterozygotes (e.g., AB blood type).
3. **Epistasis:** Expression of one gene depends on **another gene** (modifier genes).

Linkage & Recombination

- **Linked genes:** Genes on the **same chromosome**, do **not assort independently**.
- **Linkage groups:** Equal to the **number of homologous chromosome pairs** in a species.
- **Genetic Map:**
 - 1 map unit (centimorgan, cM) = 1% recombination frequency.
 - **Close loci:** Low probability of crossover.
 - **Distant loci:** High probability of crossover.

Important Concept

- Number of genes in a cell **far exceeds number of chromosomes** → chromosomes carry **many genes**.

Quick Tip for Exam

- Use **Punnett squares** for monohybrid/dihybrid crosses.
- Remember: **Linkage reduces independent assortment**, but recombination can separate linked genes.
- Centimorgan = 1% recombinant frequency.

Clinical Genetics Overview

Pedigree Analysis

- Pedigrees are used to **track inheritance** of traits or disorders in families.
- **Symbols:**
 - Male: ♂ (square)
 - Female: ♀ (circle)
 - Affected: filled shape
 - Carrier: half-filled shape

Modes of Inheritance

1. Autosomal Dominant Disorders

Features:

- Manifested in **heterozygous state**.
- **At least one parent** of the affected child is also affected.
- Both males and females transmit the condition.
- Each child has a **50% chance** of inheriting the disease.

Additional Notes:

- **Reduced Penetrance:** Some individuals inherit mutant allele but appear normal.
- **Variable Expressivity:** Severity varies (e.g., neurofibromatosis type 1: café-au-lait spots → tumors).
- **Delayed Onset:** Some diseases manifest later in life (e.g., Huntington's).

Example: Neurofibromatosis type 1

2. Autosomal Recessive Disorders

Features:

- Manifested only in **homozygous state**.
- Parents usually **unaffected carriers**.
- Siblings have **25% chance** of being affected.
- Often results from **consanguinity**.
- Expression tends to be **uniform**.

Example: Phenylketonuria (PKU)

- Deficiency in **phenylalanine hydroxylase (PAH)**.
- Leads to **phenylalanine accumulation** → converted to phenylketones.
- Untreated PKU → mental retardation, seizures, musty odor.
- **Dietary management** (low phenylalanine, high tyrosine) can almost completely control the disorder.

3. X-Linked Disorders

General Features:

- All sex-linked disorders are **X-linked**.
- **Y-linked disorders** are extremely rare.
- **Transmission:** Heterozygous females transmit to **sons** (hemizygous).
- Affected males rarely transmit disease to sons; all daughters become carriers.
- **X-inactivation** leads to partial expression in females.

X-Linked Dominant: Rare.

- Affected male → all daughters affected, sons unaffected.
- Affected female → 50% of sons and daughters affected.

X-Linked Recessive: More common.

- Males affected; females usually carriers.

4. Multifactorial Disorders

Features:

- Trait influenced by **multiple genes + environmental factors**.
- Examples: diabetes, height, blood pressure, hair color.
- **Threshold effect:** Disorder manifests when enough gene variants + environmental triggers exist.
- **Recurrence Risk:**

- First-degree relatives: 2–7%
- Identical twins: 20–40%

Cytogenetic Disorders

Karyotyping

- Visual representation of **chromosomes in metaphase**, stained and arranged by decreasing length.
- Chromosomal abnormalities in 1 in 2000 newborns; 50% of first-trimester abortions.

Chromosomal Nomenclature

- **Polyplody**: Exact multiples of haploid number ($3n$, $4n$).
- **Aneuploidy**: Non-multiples of haploid number → monosomy ($2n-1$) or trisomy ($2n+1$).
- Caused by **nondisjunction** (meiosis I or II).

Autosomal Cytogenetic Disorders

Down Syndrome (Trisomy 21)

- Extra chromosome 21 → mostly maternal origin.
- Risk increases with **maternal age**.
- Clinical Features:
 - Single palmar crease (simian crease)
 - Epicanthic folds
 - Flat facial profile
 - IQ ~25–50
 - Cardiac malformations

Sex Chromosome Disorders

Klinefelter Syndrome (47 XXY)

- Extra X chromosome → males.
- Features: microorchidism, hypogonadism, gynecomastia, infertility, mild mental retardation.
- 15% mosaic cases.

Turner Syndrome (45 X)

- Monosomy X → females.
- Features: short stature, lymphoedema, broad chest, reproductive sterility, obesity.
- 43% mosaic cases.

Key Pedigree Examples

- **PKU:** Autosomal recessive
- **Neurofibromatosis:** Autosomal dominant
- **X-linked disorders:** Hemophilia, Duchenne muscular dystrophy

Mendelian Recap

- **No blending:** Traits are particulate.
- **Dominant vs Recessive:**
 - Dominant trait expressed in F1 if present.
 - Recessive trait may skip a generation.
- **Gametes:** Carry only **one allele per gene**.
- **Mendel observed:** Wrinkled seed trait never expressed in F1, reappeared in F2
- 1.1 region shifts to allow template strand entry.